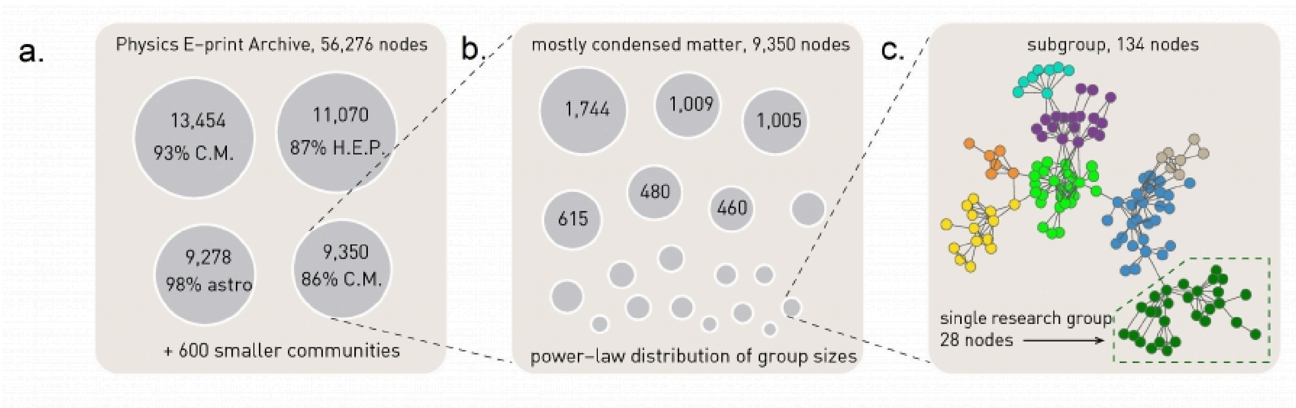
L7: Hierarchical Modularity Through Recursive Community Detection



The Greedy Algorithm. (Image 9.17) from networksciencebook.com

Another approach to investigate the presence of hierarchical modularity is by applying a community detection algorithm in a recursive manner. The first set of discovered communities correspond to the top-level, larger communities. Then we can apply the same algorithm separately to each of those communities, asking whether there is a finer resolution community structure within each of those top-level communities. The process can continue until we reach a level at which a given community does not have a clear sub-community structure.

To illustrate this process, the visualization above refers to a collaboration network between about 55,000 physicists from all branches of physics who posted papers on arxiv.org. At the top-level *(part a)*, the greedy modularity maximization algorithm detects about 600 communities and the modularity is M=0.713. The largest of those communities include about 13,500 scientists and 93% of them publish in condensed matter *(C.M)* physics. The second-largest community consists of about 11,000 scientists and 87% of those scientists publish in High Energy Physics *(HEP)*.

If we apply the same community detection algorithm to the third-largest community, we identify a number of smaller communities, and the modularity of that community assignment is 0.807 *(part b)*.

If we continue at a third level, applying the same algorithm to a community of 134 nodes we reach a point at which we identify research groups associated with scientists at different universities *(part c).*